SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I S.V. DILIP *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.V.

8.5

level - 2

CO1 - AT3

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1) Smart Home Automation System

Understanding of Concepts.

A smart home automation system controls lighting, temperature, and security using sensors and user preference. It responds to real-time event such as motion detection room temperature and time of day while learning user habits.

Explanation.

*) Simple Reflex Agent. Turns light on when motion is detected and off when no motion is detected.

*) Model-Based Agent. maintains information about room occupancy and temperature to make better decisions.

*) Goal-Based Agent: work towards goal such as maintaining comfort, reducing energy consumption, and ensuring security.

Application.

A hybrid approach combining model-Ba and utility-Based Agents is the most effective the model-based agent remembers user preference, while the utility-based agent choos action that maximize comfort and minimize.

Conclusion.

A hybrid intelligent agent provides better comfort, energy efficieny and security by adapting to changeing conditions and learning user behaviour over time.

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2) Robot in an unknown maze.

Understanding of concepts.

A robot must find the exit in an unknown maze without prior knowledge. It uses search algorithm to explore differet paths.

Explanation.

* Uniform cost search (UCS): Expands the path with the lowest total cost and guarabs the least-cost solution.

* Iterative deepening search (IDS):

Perform repeated depth-limited searching with increasing depth until the goal is found.

Application.

UCS is suitable when movement cost are different, while IDS is useful when memory is limited. for an unkown maze UCS with a Goal-Based Agent is the best choice beacuse it find the optimal path efficiently.

Conclusion.

A Goal-Based Agent using UCS provides accurate and cost-effective navigation while IDS is useful when memory efficiency is more important.

3) Smart traffic management system

Understanding of concepts.

A smart traffic management system controls traffic light based on vehicle density pedestrian movement, and weather conditions to reduce congestion.

Explanation.

* Simple Reflex Agent. changes signals based only on current traffic.

* Model - Based Agent : ~~uses previou~~ uses previous traffic data to predict congestion.

* Goal - Based Agent : Selects signal timings. that maximize traffic efficiency and minize delays.

Application.

A hybrid model-Based and untility-Based Agent is that best solution. It learns traffic patterns, predicts congestion and adjusts. signal timing according to real-time.

Conclusion.

The hybrid approach improve traffic flow, redues trave time and increase road safety by making intelligent and adaptive decision.